Table 1: Clustering quality results for the BANK-40K dataset.

					Lloyd		Cost					$\text{Gap}^{\text{SFC}}_{\text{Cost}}$ [%]				Balance					
k	λ	Target	p	q	Cost	Bal.	MIQCP	SetVars	MPFC	Flow	SFC	MIQCP	SetVars	MPFC	Flow	Lloyd	MIQCP	SetVars	MPFC	Flow	SFC
2	0.01	0.47	67	144	2,396	0.46	-	2,399	**2,397**	**2,397**	2,426	-	-1.12	-1.18	-1.18	✗	-	✓	✓	✓	✓
2	0.10	0.42	335	792			-	**2,396**	**2,396**	**2,396**	2,536	-	-5.50	-5.50	-5.50	✓	-	✓	✓	✓	✓
2	0.30	0.33	126	383			-	**2,396**	**2,396**	**2,396**	2,410	-	-0.56	-0.56	-0.56	✓	-	✓	✓	✓	✓
5	0.01	0.47	375	806	1,326	0.10	-	1,515	**1,389**	1,402	1,831	-	-17.26	-24.15	-23.43	✗	-	✓	✓	✓	✓
5	0.10	0.42	291	688			-	1,400	**1,380**	1,390	1,718	-	-18.47	-19.67	-19.09	✗	-	✓	✓	✓	✓
5	0.30	0.33	176	535			-	1,394	**1,361**	1,365	1,590	-	-12.33	-14.45	-14.18	✗	-	✓	✓	✓	✓
10	0.01	0.47	187	402	824	0.09	-	1,160	**951**	980	1,525	-	-23.95	-37.65	-35.75	✗	-	✓	✓	✓	✓
10	0.10	0.42	85	201			-	1,093	**930**	950	1,370	-	-20.25	-32.11	-30.67	✗	-	✓	✓	✓	✓
10	0.30	0.33	124	377			-	990	**890**	901	1,285	-	-22.98	-30.73	-29.92	✗	-	✓	✓	✓	✓
15	0.01	0.46	415	893	678	0.07	-	1,045	**810**	841	1,488	-	-29.76	-45.54	-43.45	✗	-	✓	✓	✓	✓
15	0.10	0.42	406	961			-	925	**786**	813	1,379	-	-32.96	-43.01	-41.07	✗	-	✓	✓	✓	✓
15	0.30	0.33	208	633			-	881	**741**	759	1,147	-	-23.23	-35.38	-33.86	✗	-	✓	✓	✓	✓
Average												-	-17.36	-24.16	-23.22						

(-) no feasible solution; (**bold**) best result; (underline) 2nd best result; (✓) target balance attained; (✗) target balance not attained

Table 2: Running times for the BANK-40K dataset.

					CPU_{tot} [s]					CPU_{avg} [s]			$\text{Gap}^{\text{SFC}}_{\text{CPU}_{\text{avg}}}$ [%]		Avg. iterations	
k	λ	Target	p	q	MIQCP	SetVars	MPFC	Flow	SFC	MPFC	Flow	SFC	MPFC	Flow	MPFC	Flow
2	0.01	0.47	67	144	limit	limit	2,105.91	176.85	limit	21.06	1.77	37.41	-43.70	-95.27	4.19	4.49
2	0.10	0.42	335	792	limit	limit	2,023.17	178.50	limit	20.23	1.79	147.55	-86.29	-98.79	4.26	4.55
2	0.30	0.33	126	383	limit	limit	1,971.85	178.99	limit	19.72	1.79	58.09	-66.05	-96.92	4.26	4.56
5	0.01	0.47	375	806	limit	limit	limit	306.56	limit	100.16	3.07	182.74	-45.19	-98.32	7.44	4.85
5	0.10	0.42	291	688	limit	limit	limit	328.54	limit	85.86	3.29	131.41	-34.66	-97.50	6.93	5.28
5	0.30	0.33	176	535	limit	limit	limit	408.30	limit	87.99	4.08	75.72	16.20	-94.61	7.51	6.81
10	0.01	0.47	187	402	limit	limit	limit	367.43	limit	362.67	3.67	95.78	278.65	-96.17	11.20	3.12
10	0.10	0.42	85	201	limit	limit	limit	454.45	limit	278.47	4.54	44.69	523.11	-89.84	10.46	4.13
10	0.30	0.33	124	377	limit	limit	limit	600.92	limit	278.06	6.01	57.32	385.10	-89.52	11.46	5.82
15	0.01	0.46	415	893	limit	limit	limit	389.36	limit	452.15	3.89	201.00	124.95	-98.06	8.25	2.12
15	0.10	0.42	406	961	limit	limit	limit	431.87	limit	361.42	4.32	174.17	107.51	-97.52	9.00	2.47
15	0.30	0.33	208	633	limit	limit	limit	593.17	limit	330.00	5.93	86.85	279.97	-93.17	9.00	3.80
Average													119.97	-95.47	7.83	4.33

(limit) time limit of 3,600 seconds reached

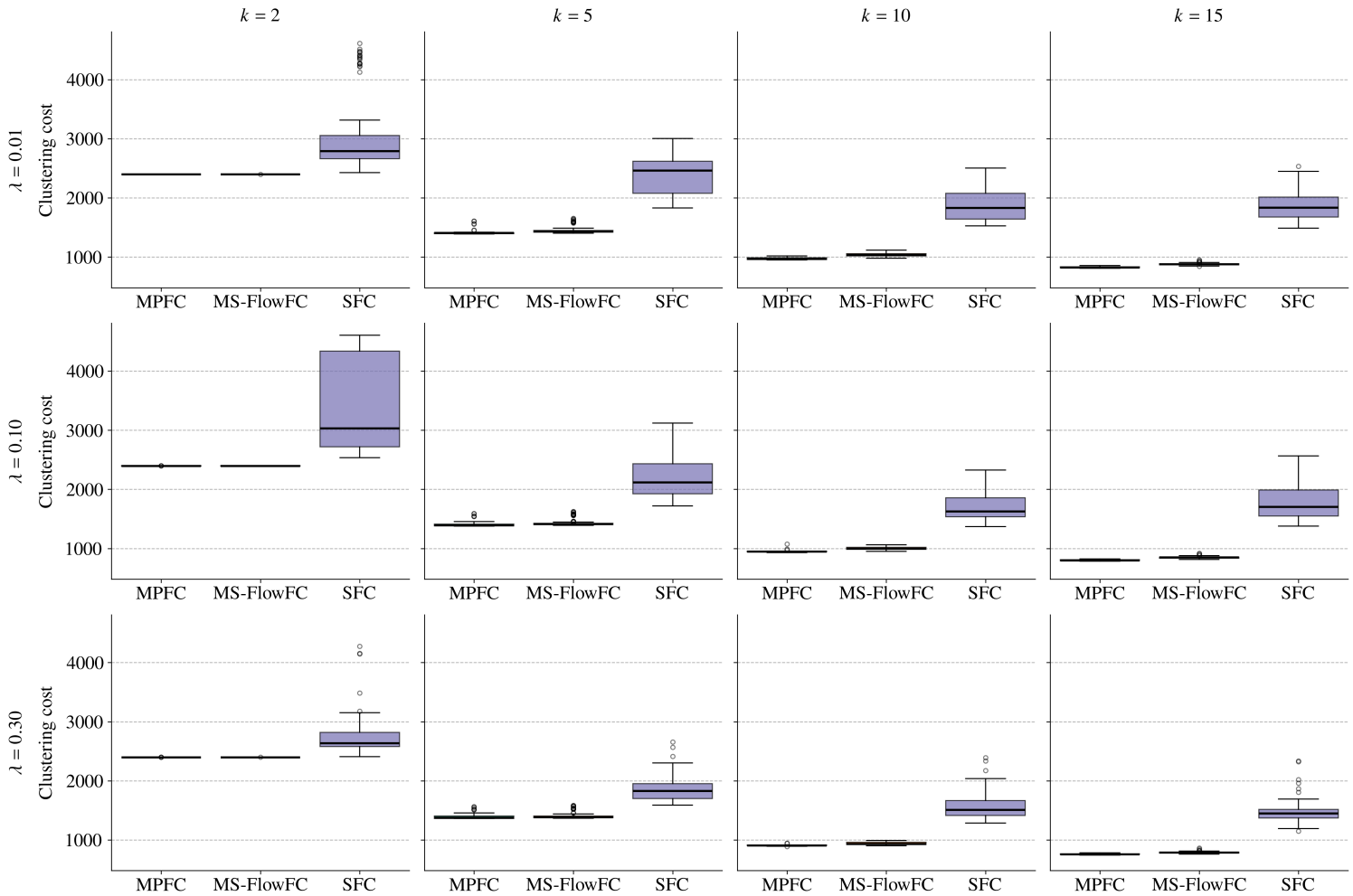

Figure 1: Boxplots of clustering costs for solutions obtained with different random seeds on the BANK-40K dataset, across different methods, numbers of clusters, and tolerance values.